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Paper Code : ESC(CSBS) 502/PCC-AI 502/PCCAIML 502 Machine Learning

UPID : 005937

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What is Regression?
- (ii) Which algorithm is used for solving temporal probabilistic reasoning?
- (iii) Agglomerative clustering follows a _____ approach.
- (iv) What are the possible values of the variable?
 - A. Possible states of the world
 - B. Literals
 - C. Variables
 - D. Discrete variable
- (v) How is the state of the process, described in HMM?
- (vi) What is the key principle used by the Apriori algorithm to reduce the number of candidate itemsets?
- (vii) Divisive clustering follows a _____ approach.
- (viii) _____ and _____ are the techniques of keyword normalization.
- (ix) What is an auto-associative network?
- (x) _____ is an example of a deterministic algorithm.
- (xi) Which of the following is not considered a level of knowledge in language understanding?
 - A. Empirical
 - B. Phonological
 - C. Syntactic
 - D. Semantic?
- (xii) The action 'STACK(A, B)' of a robot arm specifies to _____.
 - A. Place block B on Block A
 - B. Place blocks A, B on the table in that order
 - C. Place blocks B, A on the table in that order
 - D. Place block A on block B

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What is Backpropagation? [5]
3. Explain the concept of Elbow plot. Provide an illustrative example. [5]
4. Explain One-Hot Encoding and Label Encoding. Does the dimensionality of the dataset increase or decrease after applying these techniques? [5]
5. What are the applications of Naive Bayes? Is Naive Bayes is a discriminative classifier or generative classifier? [5]
6. What is fuzzy c-means clustering? What do you understand about affinity propagation? What do you understand by the term silhouette coefficient? [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. Find the means of X and Y variables and the coefficient of correlation between them from the following two regression equations: [15]

$$4X - 5Y + 33 = 0$$

$$20X - 9Y - 107 = 0$$
8. (a) Explain the core idea behind the Gradient Boosting algorithm. [5]

- (b) What role does the loss function play in Gradient Boosting? [5]
- (c) List and briefly explain three common parameters in Gradient Boosting that help prevent overfitting. [5]
9. (a) What is image tagging used for? Describe the stages of image tagging? [5]
- (b) What is tagging in machine learning? [5]
- (c) How important is Machine Learning? [5]
10. Describe the step-by-step process of FP-tree construction, where
Input: A transaction database DB and a minimum support threshold.
Output: FP-tree, the frequent-pattern tree of DB. [15]
11. (a) You run your regression on different subsets of your data, and in each subset, the beta value for a certain variable varies wildly. What could be the issue here? [4]
- (b) Your linear regression doesn't run and communicates that there is an infinite number of best estimates for the regression coefficients. What could be wrong? [2]
- (c) What do you mean by adjusted R²? How is it different from R²? [4]
- (d) What is heteroscedasticity? What are the consequences, and how can you overcome it? [5]

*** END OF PAPER ***